THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Using Euclid's subtraction algorithm for gcd$(615, 410)$, the number of subtraction steps needed is: 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 2

2. Set r = 0. While n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Starting with n = 2736, the final r is: 1

(a) 6363 **(b)** 6462 **(c)** 6372 **(d)** 6381

3. Using gcd$(60, 192) = 12$, the LCM of 60 and 192 is: 1

(a) 1020 **(b)** 972 **(c)** 960 **(d)** 948

4. To test whether 2161 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{2161}\rfloor$. The last number checked is: 1

(a) 1080 **(b)** 45 **(c)** 47 **(d)** 46

5. A counter starts at 0. For i = 1, 2, …, 56, the counter increases by 1 whenever i is divisible by 3 or by 5. The final counter value is: 1

(a) 28 **(b)** 27 **(c)** 25 **(d)** 26

6. By Euclid's algorithm, gcd$(3502, 1071)$ is: 1

(a) 19 **(b)** 16 **(c)** 18 **(d)** 17

7. Start with $n = 12$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 9 **(b)** 10 **(c)** 11 **(d)** 8

8. Using Euclid's subtraction algorithm for gcd$(1790, 1060)$, the number of subtraction steps needed is: 1

(a) 12 **(b)** 13 **(c)** 15 **(d)** 14

9. **Assertion (A):** The gcd of 36 and 60 is 12. 1

Reason (R): The gcd of two numbers is always larger than both numbers.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The number 12 has exactly 5 divisors. 1

Reason (R): The divisors of 12 are 1, 2, 3, 4, 6 and 12.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A floor of 380 cm by 171 cm is to be covered by identical square tiles, as large as possible, with no cutting. Use Euclid's subtraction algorithm to find the tile side and the number of tiles. 2

12. Find the gcd of 336 and 138 by Euclid's subtraction algorithm and use it to find their LCM. 2

13. Find the gcd of 260, 560 and 170 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 2

14. A floor of 215 cm by 301 cm is to be covered by identical square tiles, as large as possible, with no cutting. Use Euclid's subtraction algorithm to find the tile side and the number of tiles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Trace: count = 0; for i = 1 to 5: for j = i to 5: count = count + 1. Find the final count and explain the pattern. 3

16. Trace: total = 0; for i = 1 to 9: if i is a multiple of 3, add i^2 to total. List the numbers added and the final total. 3

17. Start with n = 21. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

OR

Trace: count = 0; for i = 1 to 12: for j = i to 12: count = count + 1. Find the final count and explain the pattern. 3

18. Trace: total = 0; for i = 1 to 6: if i is a multiple of 3, add i^2 to total. List the numbers added and the final total. 3

OR

Start with n = 19. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A floor measures 84 cm by 36 cm. Dev wants to cover it exactly with the largest possible square tiles, so he finds $\gcd(84, 36)$ using Euclid's subtraction method: repeatedly replace the larger number by the difference until the two numbers are equal. 5

(i) What is the pair after the first subtraction?

(ii) What is the pair after the second subtraction?

(iii) What is the gcd?

(iv) State the side of the largest square tile and the number of tiles needed.

20. Find the largest number that divides both 693 and 528 exactly, using Euclid's subtraction algorithm. Hence list all the common divisors of the two numbers. 5

OR

Find the gcd of 540, 870 and 159 by applying Euclid's subtraction algorithm twice: first to two numbers and then to the result and the third. 5

END OF QUESTION PAPER — Total: 40 Marks